Natural Language Processing E25
Experimental Analysis of Biases in Large Language Models and Metrics

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Abstract

This paper examines the existence of biases in popular large language models as well as biases in popular evaluation metrics. It will specifically examine models and metrics created with the purpose of generating and evaluating summaries. The evaluations are done on the TL;DR dataset as it provides a large amount of text and summaries divided into 29 different categories. Comparisons were done on summaries from 5 different models which were evaluated using 5 different metrics. The results where compared across all different categories, and the summaries from categories that deviated the most from the mean were subjected to human evaluations and a case study. The study found several different biases including an omission bias towards texts where important aspects of the story only account for very small parts of the text. Additionally, an evaluation bias was found in the NOIR metric, as its focus on capturing semantic meaning of the original post, does not allow it to detect important missing details. Furthermore, a bias was found towards the topics of "breakups" and "offmychest", a hallucination bias was found in the category of "tifu" and an omission bias was found in the "loosit" category. In conclusion, the study uncovered several pitfalls that one must be aware of when choosing which methods to evaluate one's generated summaries as well as when deciding what summarization models to use.

1 Introduction

The use of large language models (LLMs) has seen a drastic increase during the last few years due to their very efficient and versatile problem solving abilities. One task, where the efficiency of LLMs creates especially large advantages, is the task of summarizing documents as this can save humans large amount of time by allowing them to read short summaries instead of long documents. It is crucial that the summaries capture all important details of the text as well as gives a short and correct summarization. Thus, we need two things: effective models for summary generation and effective metrics for measuring the quality of the summaries. Many different models and metrics have been proposed in order to create summaries and measure their quality. The models and metrics should be able to create the summaries and measure the quality without being biased on the type of text that is processed.

This paper explores five different model architectures for summary generation: BART-small [1], BART-large [2], T5-small [3], T5-large [4], pegasus-large [5], as well as five different metrics for evaluating the quality of the summaries: BERTScore [6], BLEURT [7], NOIR [8], BLEU [9], ROUGE [10]. We investigate these models to find biases in the metrics and biases based on the type of text that must be summarized in order to find types of texts on which the models or metrics perform particularly well or poor.

2 Related work

Since we focus on evaluation of summarization tasks, rather than to summarization-self, we are mentioning only few works about text summarization. Authors of [11] from September 2024 showcase an overview over state-of-art techniques, challenges, and possibilities for future research within abstractive summaries. They underline the advantages of abstractive summaries. Since they can generate new phrases, they can capture main concepts from the original input better than extractive techniques which "cherry-pick" statements extracted directly from the input text [11, p. 1, 3]. A recent work from April 2025 [12] presents a detailed survey of techniques, system architectures, evaluation methods, and datasets.

2.1 Evaluation of Summarization models

The already mentioned article [12, p. 15-18] shows a nice overview over evaluation methods for abstractive tasks. Both automatic and manual evaluation are presented in a very structured way. While this work presents a general overview, the article [13] shows some concrete examples of both models and metrics for diverse datasets. For their analysis, they are using four leading pre-trained LLMs models, e.g. BART and FLAN-T5, and evaluate them with 5 metrics which include ROUGE and BERTScore. The conclusion of their experiments is that the tested LLMs are able to produce "coherent, informative, and concise summaries" [13, p.4]. However, there is a space for improvement considering factual consistency and syntactical accuracy.
[14] talks about abstractive summarization task and their evaluation techniques. The authors demonstrate an example usage of the traditional evaluation methods ROUGE and BERTScore, as well as using an LLM (concretely GPT-4) as an evaluator. Generally, the focus of evaluation on summaries includes different quality metrics, e.g. coherence, conciseness, readability, and context [14]. The authors state their concerns about ROUGE and BERTScores, as these two (as well as other metrics) traditional evaluation metrics are reliable and concrete, they may not always correlate with with the actual quality of summaries, i.e. judged by human evaluation.
ROUGE (Recall-Oriented Understudy for Gisting Evaluation) primarily evaluates based on overlap of words in the generated text and the label text. This n-gram overlap approach does not allow the metric to capture the true essence of a well generated summmary [14]
On the other hand, BERTScore and the leveraging of contextual embeddings from the model BERT allows this metric to compare the embeddings of of the predicted and reference (label) sentence. Therefore it can capture the semantic similarities better that traditional n-gram metrics [14]. However, BERTScore might still miss nuances and high-level concepts which could be noticed by a human evaluation [14].

2.2 Biases

Bias in NLP does not always have a clear definition and it is an overloaded term [15]. Therefore, we will go over some examples of bias in ML and how they can be defined and move towards bias focused on summarization tasks. Our analysis is not oriented towards social impact bias but towards topic bias.
When we think about bias in ML, we focus on biases which might raise some ethical concerns, because output of ML models can lead to unfair outcomes and reinforce discrimination against, for example, non-white females [16]. The taxonomy of such biases can be categorized into four groups: Data Bias, Algorithmic Bias, Evaluation Bias, and Deployment Bias [16]. I. Gallegos et al. in [17] focus their research on "bias evaluation and mitigation techniques for LMMs" [17, p. 1] to avoid harmful social biases which some LLMs can cause. Another type of bias is metric blindness, which is an evaluation metric introduced in [18]. Metric blindness occurs when metrics hides subgroup failures, which as an example could happen if a metric did not punish hallucinations in the generated texts.

We move towards literature focused more to summarization tasks and their biases, such as [15] where the authors state, that we would like to expect that AI generated summaries would be faithful to the input text, but summarization can suffer from biases, for example, Inclusion, Hallucination, and/or Representation. Hallucination Bias defined as: "summary context [is] unsupported by the input" [15] is especially important in abstractive summarization and also for our analysis.

Z. Asimiyu in his article [19] describes bias in personalized summarization. The author distinguishes between bias generally in NLP and bias in text summarization. While bias in NLP "is a multifaceted issue that affects how information is selected, structured and delivered" [19, p. 3], bias in text summarization "manifests when generated summaries **systematically deviate** from balanced or representative coverage of the source context" [19, p. 3]. The biases in the context of summarization are defines as (taken from [19, p. 3], slightly rephrased):

1. **Selection Bias**, defined as systematic disproportional highlighting of certain sentences, entities, or topics.
2. **Framing Bias**, appears when the language of the summary changes the tone or intent of the input text.
3. **Omission Bias**, is defines as systematically excluding key facts, counterarguments, or alternative perspectives.

As stated above the term "bias" has a slight disambiguation. In our research, we were focusing on Metric Blindness (i.e. metrics is hiding subgroups failures), Hallucination bias (i.e. adding information which are not part of the original text), Selection Bias (i.e. disproportional highlighting of certain information), and Omission Bias (i.e. excluding facts).

3 Methods

The study was conducted using the TL;DR dataset [20], as the division of the data in subreddits[1] creates a good starting point for exploring bias in texts with different topics and styles. There are 29 subreddits with disproportional number of elements (see Appendix D and Figures 4 and 6). Therefore the experiments were done on 114 posts from each subreddit, since this is the amount of posts in the subreddit "cooking" with the fewest posts, and this ensures that we have maximum data while still preserving the equal representation of the different subreddits.

The experiments start by having all models generate summaries for all posts. The summaries are then evaluated using all 5 metrics, and these metrics are divided into subreddits and plotted in boxplots and line charts. Examples of these plots can be seen in Figure 1 and 2 and all of them in Appendix A. Then these plots were examined and compared across each metric and each model in order to find subreddits that either consistently scored higher or lower than the rest on a given metric, or find subreddits of which a given model consistently produces worse or better summaries. Case studies were conducted for all subreddits that were found to perform better or worse. These case studies were done by randomly choosing a subset of the generated summaries and doing a human evaluation of the summaries. The "golden annotations" are provided by 3 humans (the members of the group) that evaluated the summaries on overall quality, which includes the coherence of the text, the length and the how much of the semantic content from the original post is retained in the summary. We are aware of the subjectivity of the human annotations, and have been cautious when evaluating the summaries in order to keep it as objective as possible. However, the quality of a summary will be somewhat subjective as humans can disagree on which qualities makes a good summary. The golden annotations were then compared to the relevant metrics in order to see correlation between the metrics and the golden annotations. The correlation was measured using the Pearson correlation coefficient [21], which measures the linear correlation between two datasets. Therefore, the Pearson correlation coefficient is well suited for our measurements, as they do not follow the same scale, but as one metric increases the other metrics should ideally increase as well because they are all measuring the quality of the summaries. Finally, the individual feedback on each of the human evaluated summaries were examined to find causes for the better or worse summaries.

The models that were evaluated are: BART-small [1], BART-large [2], T5-small [3], T5-large [4], PEGASUS-large [5]. These models have been chosen based on their popularity and general high performance, as the study aims to find biases and provide insights into the use of different and commonly used models. The models were be evaluated on the following five metrics: BERTScore [6], BLEURT [7], NOIR [8], BLEU [9], ROUGE [10]. The metrics had been chosen since they are

[1]By "subbreddids" we mean subsets from Reddit with diferent topics, e.g. a subreddit called "relationships"

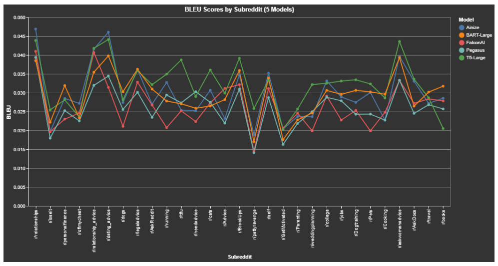


Figure 1: BLEU overview

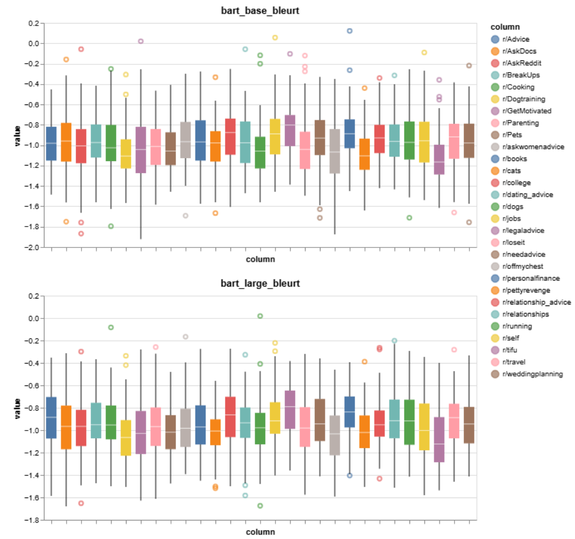
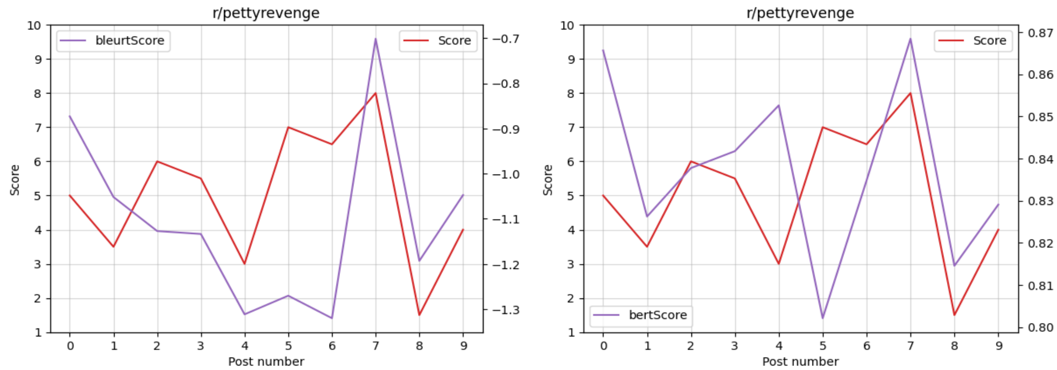


Figure 2: BLEURT metrics for BART-base and BART-large

a collection of current state of the art measurements, such as ROUGE, BLEURT, and BERTScore, as well as newer approaches such as NOIR which aims to be length-aware and distinguishes from the other metrics by not requiring human annotations. We also chose to use the BLEU score as it measures a lexical similarity which can be useful for extractive summarization methods. Thus the BLEU score has the potential of providing insights even though it was created for machine translation.

4 Analysis

Our analysis has found multiple biases in the models in general and in specific metrics. First of all, the analysis showed lower scores across almost all metrics on all models for the subreddit "pettyrevenge". This disadvantage towards summarizing posts from the "pettyrevenge" subreddit was especially prominent for the BERTScore [6] and the BLEURT score [7]. Therefore, a human evaluation of a subset of the generated summaries was conducted and compared to the results of BERTScore and BLEURT score. This comparison can be seen in Figure 3. As the plot shows, the human evaluations and the metrics tend to agree on which summaries are good and which are bad. This conclusion is backed by the Pearson correlation coefficient which was 0.25 and 0.31, respectively. This indicates that there is a positive correlation between the human evaluations and the metrics, even though the correlation might not be very strong. This indicates that the bias might originate in models rather than in the metrics. Looking at the reasons that were provided for the bad human evaluations, the most common response is that the summaries miss the actual revenge part of the story. The reason this happens is that most stories about petty revenge consist mainly of context as to why the revenge was needed. The revenge part of the story is, in most cases, only a very small part at the end. This causes the models to miss this very crucial part of the story as it only makes up a very little part of the actual story. In addition, the study showed that this error was less prominent in BART-large [2] and T5-large [4] models. The reason why is most likely that the increased size of the models allows the model to retain more details of the story which enables the model to remember and include the tiny revenge part in the summary. This fact is further supported by the fact that the NOIR [8] metric did not show any bias towards the "pettyrevenge" subreddit. As NOIR focuses on length and semantic retention of a summary, it does not notice that the summary is missing a small important detail at the end the story. The reason is that the overall semantic meaning of the story is close to covered by the summary as it covers the entire story except for the last line explaining the revenge. This means that NOIR measures that the majority of the semantic meaning of the summary is kept, and therefore the NOIR score does not capture the missing detail. In conclusion, this shows that there is actually an omission bias in the NOIR metric, as it does not handle stories well where small parts of the story are crucial to the summary. Other metrics, like the BERT score, that also aim to evaluate summaries based on semantic retention use a human generated summary as a reference point. This allows the metric to notice the missing detail, as this is provided by the human in the given reference summary.



(a) BLEURT Scores

(b) BERTScores

Figure 3: Comparison of human evaluation, BLEURT and BERT scores on summaries from the subreddit "pettyrevenge". The Human scores have been scaled down to show how movement in the scores correlate.

The study also found that the NOIR metric generally had lower score on summaries in the subreddits "offmychest" and "breakups". In the case study of those two subreddits, the human evaluations showed to be quite similar to the evaluation of the NOIR metric with a Pearson correlation coefficient of 0.52. The fact that the human evaluations and the NOIR scores tend to agree on the quality of a summary points to a bias in the models. The case study did not reveal any specific reasons for this bias, however, the human feedback about the generated summaries was that they in general left out too much context or the essence of the story. This would explain why the NOIR score is low, as it measures the semantic meaning of the story and the generated summary, and if the summary leaves out too much context the semantic meaning will not quite be the same. Additionally, as NOIR is a length-aware quality metric, we tested if the length of the summaries had an impact on the evaluation. However, the length of the summaries in the subreddits "offmychest" and "breakups" were only around 3% percent longer than average, and the posts where approximately 5% longer than average. Thus, there is no significant impact from the lengths of the summaries, as they were only slightly longer than average and the posts were as well. In conclusion, the study found a bias across all models when generating summaries in the subreddits "offmychest" and "breakups", as the summaries left out too much of the context of the stories. However, the case studies did not uncover any real reasons as to why the models would have this bias on these subreddits.

Throughout the study, it was also significantly noticeable in our analysis, that the generated summaries by the models have a tendency to do well in the beginning of the summary, but suddenly stop in the middle of the text and missing a huge part of the narrative. This was seen especially in the subreddits that performed the worst according to the metric evaluations and human annotations. For example, two of the worst performing subreddits in the dataset is called "tifu"[2] and "loseit"[3]. In many of the models generating summaries, the models manage to catch the beginning of the post, for example what led up to the main action, but misses what actually makes it a screw up situation.
In those cases the generated summaries that missed it, often end up missing it after describing the beginning of the post and ends without describing what went wrong or the end of the story. In the end it gives an incomplete version of the story.
This tendency might indicate that some of the models may prioritize the beginning of the posts, but at the same time have a hard time to keep the narrative of the posts, especially if the story is very heavy on the narration. This causes an issue especially for posts that puts the main point at the end of the posts.

[2]"tifu" is short for "Today I fucked up".
[3]"loseit" is a subreddit about how to lose weight.

Additionally, the case study showed that there was a positive correlation between the BLEURT score and the human evaluations with a Pearson correlation coefficient of 0.32 when evaluating the subreddit "tifu". The human feedback on the summaries found that the bad summaries often contained hallucinations. This indicates that there is a hallucination bias in the subreddit "tifu". However, the stories that included hallucinations tended to have a mediocre score, which means that they did not correlate with the human annotations (Examples of hallucinations can be seen in appendix B, and summary scores for "tifu" can be seen in appendix C). This points to metric blindness and evaluation blindness [18] in the BLEURT metric, as it focuses on the semantic meaning of the story which allows it to ignore some hallucinations.

5 Conclusion

Our study found several biases in the models and the metrics. First of all, the study showed a general bias towards the subreddit "pettyrevenge". The reason for this bias is the fact that the posts in this subreddit consist mostly of context and end with a very short explanation of the revenge. The revenge part of the posts is the most important part, but the models tend to miss this part as it is only a very little part of the original post. Similarly, we discovered a bias in the NOIR metric as it did not show lower scores on the subreddit "pettyrevenge". This problem was caused by the fact that NOIR compares the semantic meaning of the summary and the post. Therefore, when the models only missed small but crucial details it would not have a significant impact on the NOIR score. Additionally, the study uncovered a general bias in the models on the subreddits "offmychest" and "breakups" where the models tend to leave out too much of the context to create a good summary. The study showed a tendency to only summarize the beginning of text in the subreddits "tifu" and "loseit", which results in the summaries missing large part of the important content of the summary. Finally, a metric blindness was found in the BLEURT metric as it tended to not penalize hallucinations in the summaries. In conclusion, the study showed that it is important to consider which model you choose depending on what type of text must be summarized. In addition, it also showed biases in certain metrics, which must be considered when evaluating summarization models.

It may be interesting to further investigate the use of LLMs as evaluators, as they might be able to learn to spot missing important details in a summary or hallucinations. In general they might be better at capturing a holistic evaluation of a summary rather than the metrics used. Additionally, it would be interesting to extend the research to combine different metrics in order to see if the combination of metrics could remove the biases in the individual metrics. Finally, it would be valuable to increase the amount of data, in the sense of having multiple reference summaries for each post and having more human annotators, as well as having a larger dataset.

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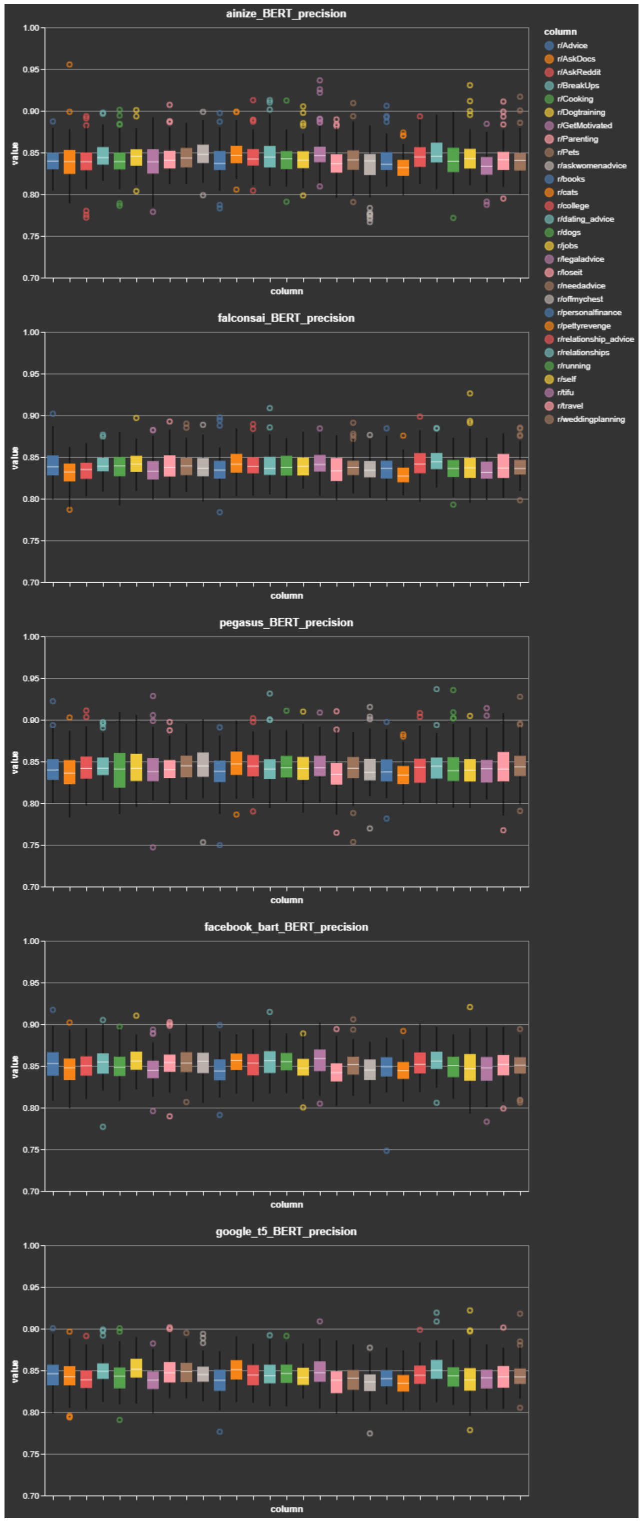
A Overview of Initial Analyses of Scores of Summaries on Subreddits

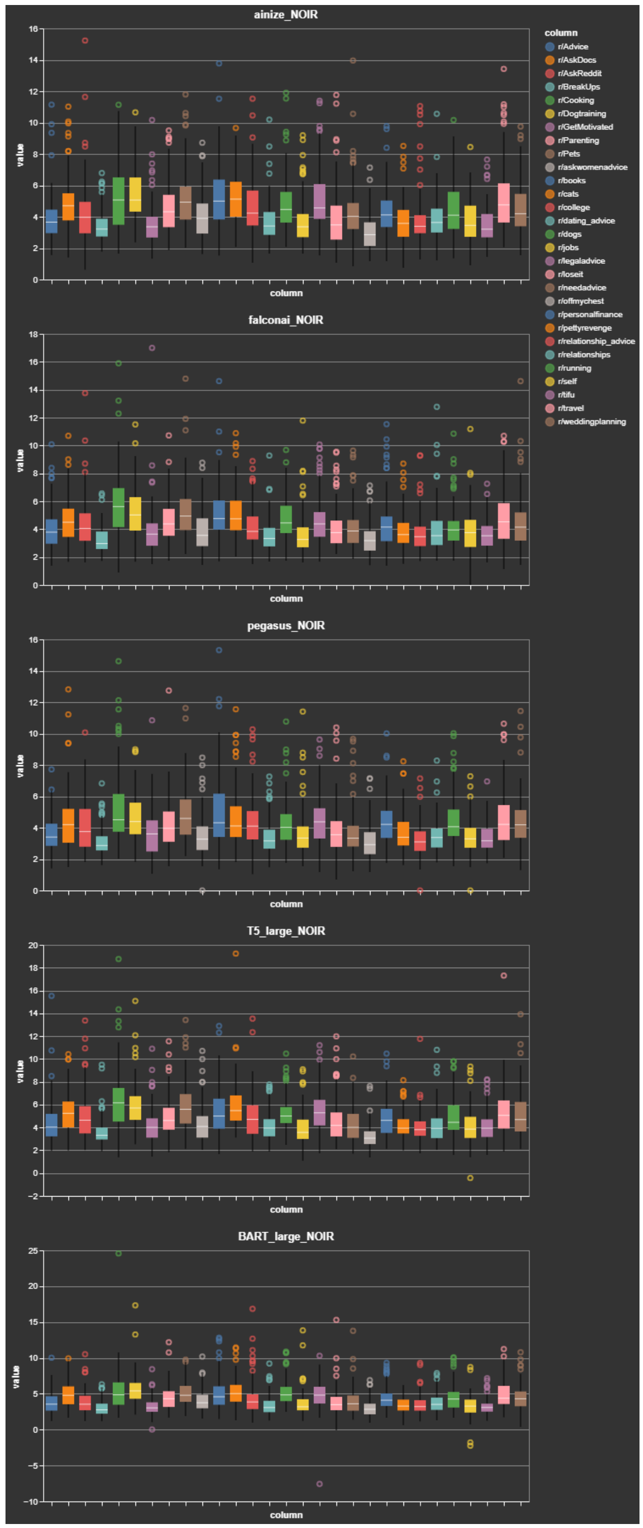
A.1 BERT Scores

This section show how the Bert scores of all summaries are distributed per subreddit. A noteworthy finding is that the subreddit "pettyrevenge" consistently scores very low using BERT.



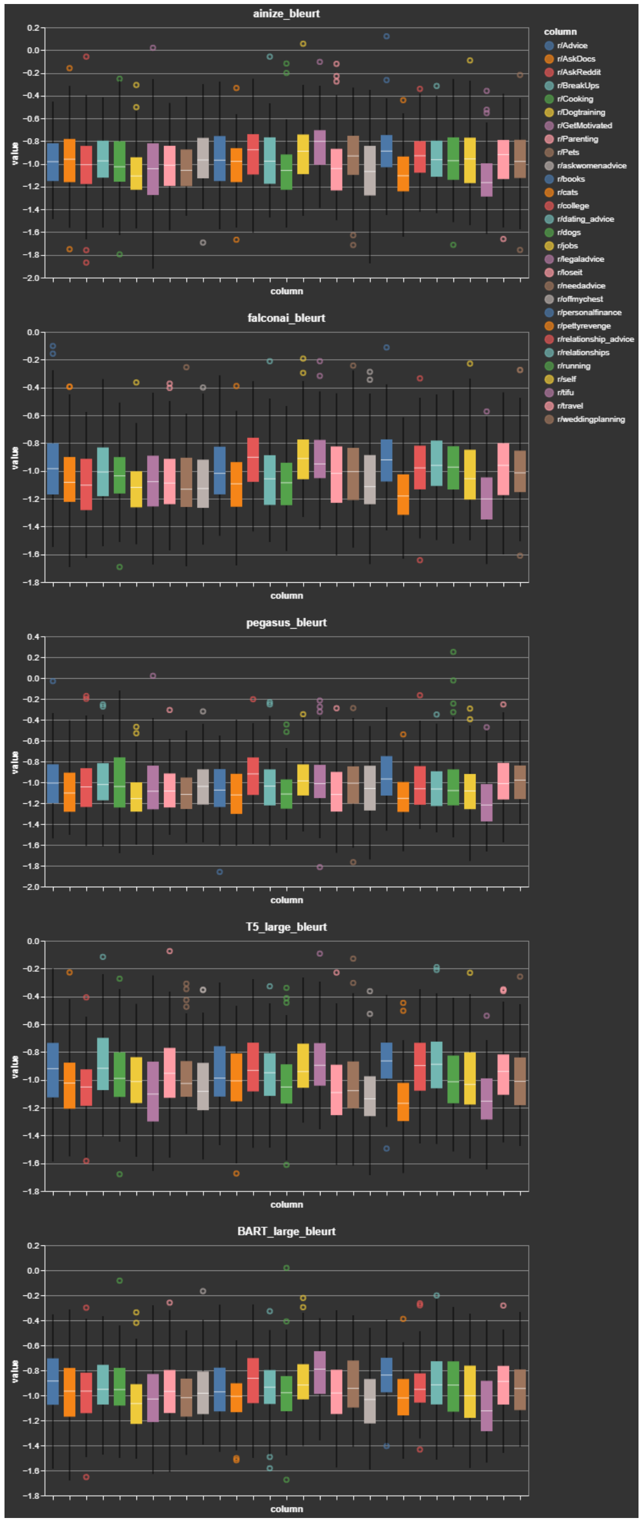
A.2 NOIR Scores

This section show how the NOIR scores of all summaries are distributed per subreddit. A noteworthy finding is that the subreddits "offmychest" and "breakups" consistently scores very low using NOIR.



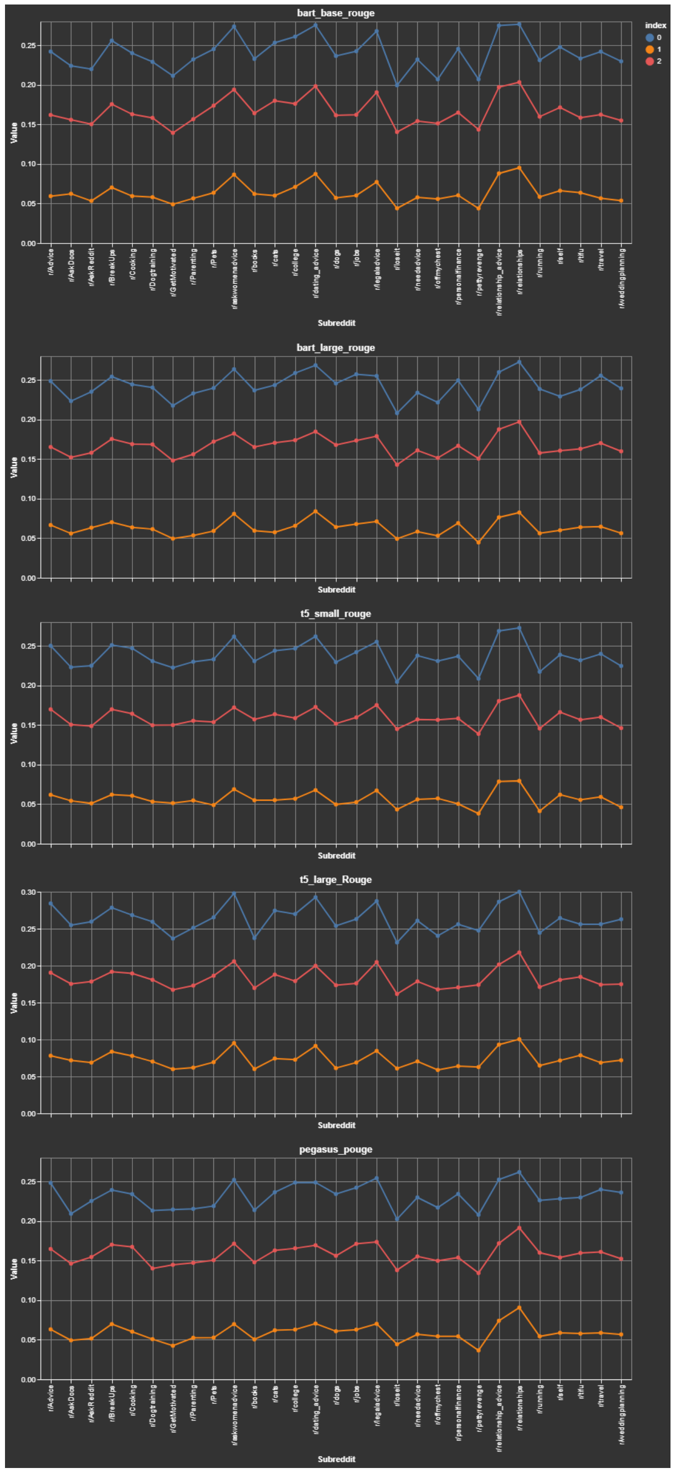
A.3 BLEURT Scores

This section show how the BLEURT scores of all summaries are distributed per subreddit. A noteworthy finding is that the subreddits "tifu" and "pettyrevenge" consistently scores very low using BLEURT.



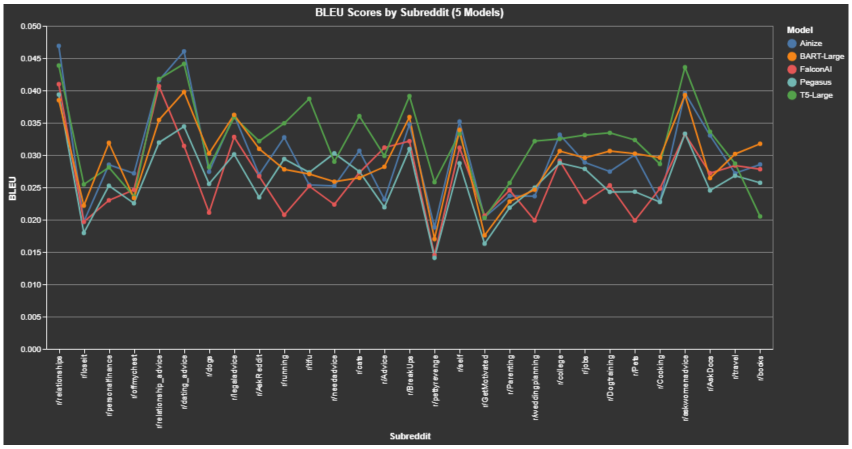
A.4 ROUGE Scores

This section show how the ROUGE scores of all summaries are distributed per subreddit. A noteworthy finding is that the subreddit "loseit" consistently scores very low using ROUGE and the subreddit "relationships" scores very high. The blue line is the ROUGE-1, red line is ROUGE-L and orange line is ROUGE-2.



A.5 BLEU Scores

This section show how the BLEU scores of all summaries are distributed per subreddit. A noteworthy finding is that the the subreddit "relationships" scores very high using BLEU.



B Examples from Case study

B.1 Hallucination Example

The following section shows a case of a summarization that includes hallucination:

POST:
"SUBREDDIT: tifu

TITLE: TIFU by wanking with hair gel (NSFW)

POST: Context: I live at home with my two parents and my brother. My parents both work and were both have been gone since Monday on their anniversary to Vegas. Anyway, my brother and I walk to the same school every morning but on Fridays, I get out at 10:30 today because I have a weird schedule that consists of an early lunch and multiple free periods. Keep in mind I have had a stuffy nose for the past week or so redditors.

Story: So, I get out of school at 10 o'clock and get home around 10:15. I hang out for about 30 minutes or so, until I realize the significance of the empty house. So, I decided to open up the spank bank. I am mildly aroused and decide that I need some lubrication. I look around my room and cannot find my lotion so I head to my parents room where I still cannot find any lotion. Checking the time, there is only about 25 minutes left to do the deed. I then go to my brother's room and find a bottle of what appears to be lotion. I rush to my room, fully naked, hoping that I will have enough time for a quickie. I turn on the content and apply lotion **generously** and the rest is history. After I am finished I put the lotion back and clean up a little.

My brother came home about 2 minutes later and obviously couldn't smell the lotion or see any of the mess. A few minutes later, I sit down to watch tv, and my balls begin to feel a little weird, but I brush it off. I fall asleep on the couch and awake three hours later in agonizing pain, my sack feeling like concrete. I think back to what happened and hobble upstairs like a penguin to my brother's room to check the lotion. Turns out, it's **hair gel**. I couldn't smell the difference when I used it because of my stuffy nose. In a desperate attempt, I jump into the shower and feel a searing pain in that area. I don't dare to look down, but when I eventually do, it feels and looks like a rock hard, sun-dried tomato.

Summary:
TL;DR:" wanking with hair gel is a tifu story about a girl with a stuffy nose . she uses hair gel to lubricate her balls and wakes up in agonizing pain . the girl with the hair gel says it's a good thing she has a spank bank . a friend of hers says she's never been happier.

B.2 Omission Example

The section shows a summarization from the "pettyrevenge" subreddit. It displays a very common issue within this subreddit, which is that it tends to leave out the very important detail of the actual revenge. In this case the summary leaves out the revenge part where the author did not let the annoying family charge their phone.

Post
"SUBREDDIT: pettyrevenge

TITLE: Contribute the most to your annoyingly loud family? Enjoy your flight with a dead battery

POST: This is happening right now. About 9 or so people, my guess friends and/or family members, were seated across and next to each other at an airport gate. They all decided to talk to one another at the most horibly loud volume. One of them, we shall all her my Droid Rage, was walking to each person and talking at the previously mentioned volume despite being in front of them, showing pictures, taking, pictures and playing music through the phone's speakers. She also would sit and play loud games, and then jump up to show them off. This went on for over an hour. I had been charging my computer and phone with the nearest outlet (taking up both because no one was around earlier.) at this point I saw her looking around somewhat frantic and then looked down at my plug and Droid Rage asked about her phone to her friend who shrugged and pointed around at other plugs but she ops out for a dead phone rather than move away alone.

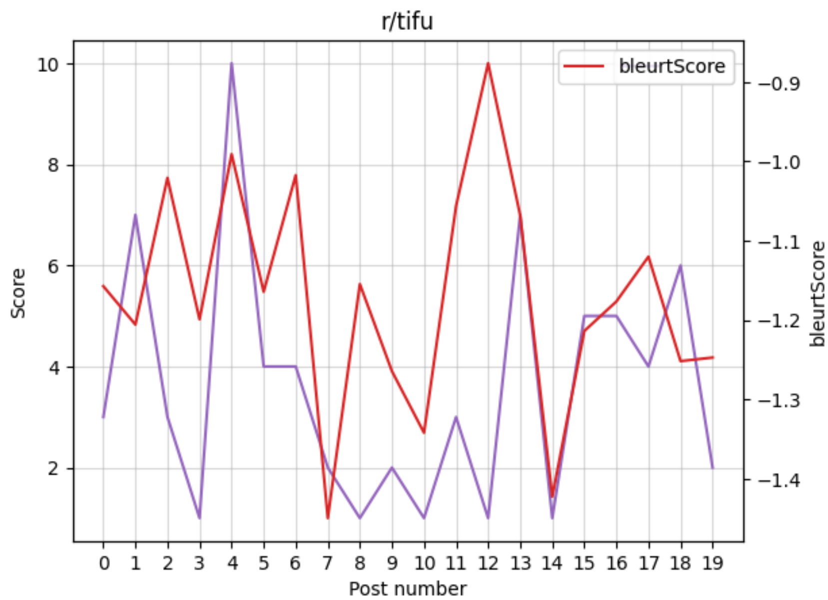
Now if I had been a nice guy I could have given up my outlet for the phone and charged it with my computer but I wanted her phone dead!

TL;DR:"

Summary
"Droid Rage was talking loudly to people at an airport gate. She was showing pictures, taking, pictures and playing music through the phone’s speakers. She also would sit and play loud games, and then jump up to show them off. This went on for over an hour. I had been charging my computer and phone with the nearest outlet. I saw her looking around somewhat frantic and then looked down at my plug. "

C Case study

This section shows the human evaluations compared to the BLERUT scores of summaries from the "tifu" subreddit.



D Amount of training data in TL;TR dataset

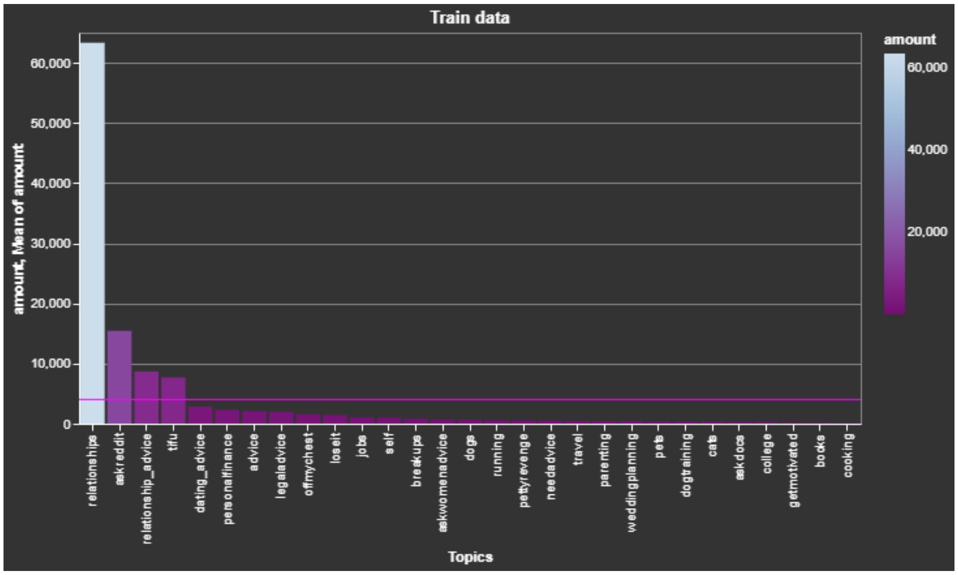


Figure 4: Number of elements in each subreddit

	Topics	amount
0	relationships	63324
1	askreddit	15440
2	relationship_advice	8691
3	tifu	7685
4	dating_advice	2849
5	personalfinance	2312
6	advice	2088
7	legaladvice	1997
8	offmychest	1582
9	loseit	1452
10	jobs	1084
11	self	1048
12	breakups	838
13	askwomenadvice	688
14	dogs	638

Figure 5: Number of elements in each subreddit

	Topics	amount
15	running	567
16	pettyrevenge	548
17	needadvice	528
18	travel	452
19	parenting	435
20	weddingplanning	433
21	pets	366
22	dogtraining	362
23	cats	324
24	askdocs	283
25	college	264
26	getmotivated	169
27	books	161
28	cooking	114

Figure 6: Number of elements in each subreddit